

MSDS-422-0 - Final Project Milestone 1-2

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(2) “Which content attributes most strongly to predict whether a post becomes viral across YouTube videos”

<https://www.kaggle.com/datasets/cyberevil545/youtube-videos-data-for-ml-and-trend-analysis>

Citation for the Kaggle Source:

CyberEvil545. 2025. *YouTube Videos Data for ML and Trend Analysis*. Kaggle.

<https://www.kaggle.com/datasets/cyberevil545/youtube-videos-data-for-ml-and-trend-analysis>

Github final project 422 repository link:

<https://github.com/mchalekson/422-final-project>

Abstract

(must be 250 words - draft)

This study investigates which content attributes best predict whether a YouTube post becomes viral, synthesizing insights from visual, textual, social, and affective features. Prior work shows that standard engagement metrics (views, likes, comments) alone incompletely capture content quality, and that incorporating comment sentiment can surface high-quality but under-ranked educational videos. Creator-level attributes, particularly subscriber count, consistently raise all tiers of engagement, whereas large upload volume depresses per-video interaction, and video age differentially affects views and likes. At the visual level, image memorability - driven by semantic distinctiveness and clear composition - predicts higher commenting and virality, largely independent of emotional intensity. Complementary research on visual atypicality indicates that visually unique content boosts engagement when it remains coherent with category norms and platform expectations, but harms engagement when perceived as deviant in high-norm contexts (e.g., informational content). Machine learning analyses of trending videos further show that post-publication engagement features, especially like counts, almost perfectly explain view variance, whereas pre-publication metadata alone has weak predictive power, underscoring the challenge of ex-ante virality prediction. Multi-view embedding approaches for micro-videos similarly find that joint modeling of visual, textual, social, and acoustic cues improves popularity prediction over single-feature models. Overall, the most influential predictors of virality appear to be (1) creator social capital (subscribers), (2) visual memorability and controlled atypicality, and (3) sentiment-weighted engagement signals, interacting with category and platform context to shape diffusion trajectories on YouTube.

Problem Statement / Research Objective(s)

Despite extensive attention to YouTube virality, there remains limited consensus on which content attributes most reliably predict viral success, particularly prior to widespread audience engagement. Existing research often focuses on post hoc engagement metrics such as views, likes, and comments, which are highly informative but offer limited explanatory power for ex-ante prediction and content design. Moreover, prior studies tend to isolate individual factors such as sentiment, visual novelty, or creator popularity, rather than examining how visual, textual, social, and contextual features jointly interact to shape diffusion outcomes. This project seeks to address this gap by systematically evaluating which content attributes most strongly predict virality across YouTube videos, using a large-scale dataset of trending and non-trending content. Specifically, the research aims to assess the relative influence of creator-level social capital, visual attributes (including memorability and atypicality), textual metadata, and sentiment-weighted engagement signals, while accounting for category norms and platform context. By integrating insights from prior empirical studies with exploratory data analysis and machine learning methods, this work aims to clarify which predictors are consistently associated with viral outcomes and which signals primarily emerge after virality has already occurred, thereby contributing to a more nuanced and realistic understanding of YouTube content popularity.

Annotated Bibliography

(Milestone 1: 6 peer-reviewed articles, each one should be summarized and annotated to no less than 300 words.)

Selin's sources -

Adam, Zitian, and Zhiyong Yang. "Breaking the Mold: How Visual Atypicality Shapes Brand-Generated Image Virality." *International Journal of Research in Marketing*, advance online publication, Elsevier B.V, 2025. <https://doi.org/10.1016/j.ijresmar.2025.07.005>.

Content Characteristics, Context, and Conditional Virality on YouTube

This research seeks to provide a more detailed analysis of why some content succeeds and other content fails in social media space. By demonstrating that different content attributes are effective depending on whether they align with the audience's expectations and category norms, we demonstrate that the novelty of an item is just one factor that influences its potential for going viral.

We use large-scale image data and machine learning based content attribute representations to show that visually unique or atypical items can generate significant amounts of engagement (in the form of likes, shares, etc.) when the content attributes are aligned with expectations that will be met. We also find that the content attributes of visually unique, semantically deviant, and compositionally unique content can be used to predict increased levels of engagement when those attributes create perceptions of entertainment or uniqueness in the eyes of the viewer. Conversely, we also find that content attributes that are perceived as being deviant and do not meet expectations, especially in categories where there are high expectations (e.g., categories that have well-established norms), can lead to decreased levels of engagement.

This conditional relationship emphasizes the role of context in determining whether or not an item will go viral. Specifically, we find that visual attributes associated with an item can be used to predict increased levels of engagement on YouTube; however, those same attributes must be examined in relation to the category of video being watched (e.g., entertainment, instructional) and platform-specific norms (e.g., video length, thumbnails). For example, a visually unconventional thumbnail could potentially create additional engagement in an entertainment category, while a similar deviation could potentially decrease engagement in an informational category.

In addition to providing insight into the nature of virality, this study also demonstrates the importance of developing models of virality that include the interaction of multiple content attributes and contextual variables, rather than using single-attribute approaches. Furthermore, we show that engagement metrics (such as views, likes, and comments) are influenced by both the quality of the content and the degree to which it meets the audience's expectations.

Overall, this research provides additional insight into the nature of virality by showing that content attributes are most likely to contribute to an item going viral when they are both unique and coherent within the context of the platform and category. Therefore, our results support predictive frameworks that utilize visual content attributes, category information, and engagement signals to model viral diffusion.

Peng, Shikang, and Wilma A Bainbridge. "Image Memorability Predicts Social Media Virality and Externally-Associated Commenting." *Computers in Human Behavior* 174 (2026): 108799. <https://doi.org/10.1016/j.chb.2025.108799>.

Image Memorability Predicts Social Media Virality and Externally-Associated Commenting

Examining what attributes of online content best predict successful online engagement (and specifically video success), this study examines the relationship of visual memorability to viewer engagement with social media content. Rather than view video success as an emotionally driven response (as many studies have) or as solely determined by algorithms used to rank videos, this study views video success as the result of how well the viewer perceives, processes and remembers the video's visual content.

Using large datasets of images and computational memorability models, the authors found that, regardless of other aesthetic qualities of the content, visually memorable content consistently resulted in greater user engagement (commenting and sharing) on social media. These results are especially important for YouTube, where video thumbnails and representative frames of a video are very important to get the first look from a potential viewer and ultimately determine whether they will view the video. In terms of visual characteristics, content that is considered visually memorable has a clear composition, is semantically distinct and has recognizable visual structures that make it more likely to be recognized and remembered by the viewer. All of these factors contribute to the likelihood that a user will interact with the content beyond simply viewing it, thus contributing to the content going "viral".

Importantly, this study also showed that memorability works independent of emotional intensity, suggesting that content can go viral without being extremely affecting or sensational. Therefore, in terms of YouTube analysis, this study demonstrates the value of including visual scale and presentation (such as resolution, framing and structural clarity) as predictive variables for video success. Since memorability itself is typically not directly measurable in all data sets, the effect of memorability is often observed through aggregated engagement metrics (views, likes, comments). This study offers theoretical support for the use of engagement outcomes as proxy measures for underlying cognitive mechanisms in predicting viral success.

Ultimately, this study further validates that the attributes of online content (category) related to visual clarity (resolution, bitrate, frame rate) and semantic distinctness (titles, descriptions, hashtags) are key predictors of viral success and provide a cognitively grounded explanation for why some YouTube videos achieve disproportionate reach and interaction.

Konukoğlu, L., & Özman, M. F. (2025). *Video duration and engagement in mathematics instruction on YouTube: An analysis of content-level moderators*. International Journal of Curriculum and Instruction, 17(3), 689–711.

<https://ijci.net/index.php/IJCI/article/view/1633>

Video Duration, Content-Level Metadata, and User Engagement on YouTube (2025)

The present research will assess how video duration interacts with user engagement metrics (i.e., views, likes, and comments) within a body of instructional math videos on YouTube. To analyze how video length predicts user engagement and how it varies across content-level categories (e.g., educational level, instructional focus, and mathematical topic), the researchers used a sample of 126 highly popular Turkish-language videos from a public database. Since all measures of user engagement were based on publicly available platform metrics, this research provides a basis for comparison with studies that use structured metadata collected from YouTube and other social media platforms (rather than proprietary audience data).

The methodology of the current research involved applying both descriptive statistics, one-way ANOVA, and Pearson correlation analyses to assess the relationships between video duration and user engagement metrics. While the researchers found strong positive correlations among their three metrics of user engagement (likes and comments were significantly correlated), they found no statistically significant linear relationships between video duration and the number of views, likes, or comments. As such, these results challenge commonly cited hypotheses that users prefer shorter videos because they are likely to have higher engagement (a hypothesis that posits that video duration can serve as a predictor of user engagement). Additionally, the results provide further evidence that video duration cannot function as a stand-alone variable for predicting user response. The results also highlight the potential for using continuous metadata variables (such as video duration) when analyzing large datasets, and suggest that while video duration itself is not a good indicator of user engagement, structural metadata (including duration and categorizations by content-level) can help provide insight into how videos were produced and designed for instruction.

Although video duration was not an independent predictor of user engagement, the researchers observed substantial variability in video length across content-level categories. In particular, the videos classified as "exam-focused" or "exercise-based" tended to be significantly longer than videos classified as "topic-based." These differences in video length appear to reflect differences in the instructional intent of each category of videos (i.e., a difference in what the videos were intended to teach students, rather than a difference in how students preferred to consume them). Furthermore, the researchers found differences in video length across branches of mathematics and educational levels, although none of these differences were statistically significant for user engagement. This finding underscores the importance of including structural metadata (e.g., duration and content-level) in the analysis of user engagement patterns; in addition to serving as indicators of user engagement, such metadata can be useful for describing how instructional videos were designed and produced.

The results of the current research support its relevance to engagement modeling with metadata, as they show that content-level attributes, such as video duration and categorizations) can be systematically analyzed alongside user engagement metrics (view counts, like counts, comment counts) using conventional statistical procedures. The current research also supports the inclusion of production-related metadata (e.g., video length and categorizations) in exploratory data analysis and feature engineering for large datasets that include variables such as video length, category labels, and engagement metrics. Therefore, the results of the current research support the view that structural metadata can serve as valid analytical inputs that provide context for understanding user engagement patterns, thereby supporting a metadata-driven approach to modeling video popularity on YouTube.

Max's sources -

Hussain et al. (2024) - Two-actor model and social capital in YouTube tourism

References

Hussain, K., Nusair, K., Junaid, M., & Aman, W. (2024). A two-actor model for understanding user engagement with content creators: Applying social capital theory. *Comput. Hum. Behav.*, 156, 108237. <https://doi.org/10.1016/j.chb.2024.108237>

Hussain et al. (2024) develop a two-actor model to explain how content creator characteristics and user responses jointly shape engagement with tourism videos on YouTube, grounding their framework in social capital theory (Hussain et al., 2024). They analyze 366 tourism-related videos across six platforms - data collected via VidIQ, TubeBuddy, and SocialBlade. From there they model three tiers of engagement - views, likes and comments - using PLS-SEM in SmartPLS 4.0. Creator-level variables include subscriber count, total uploads, and video age, which are linked to these engagement outcomes. Subscriber count has a consistently positive effect on all three engagement tiers: channels with more subscribers gain more views, likes, and comments per video, underscoring the role of accumulated structural social capital in driving interaction. In contrast, a higher total number of uploads is associated with lower engagement, suggesting diminishing marginal attention or audience fatigue as creators expand their video portfolios. Older videos accrue more views over time, but the likelihood of receiving likes declines as the videos age. The study supplements behavioral metrics with sentiment analysis of 23,933 comments using the VADER tool. Results show that 59.5% of sentiments are positive, 31.2% neutral, and only 9.3% negative, indicating a broadly favorable orientation toward tourism content on YouTube. A temporal analysis from 2010 to 2023 reveals a rising trend in qualitative engagement (commenting), interpreted as growing interest not only in consuming but also in interacting with tourism-related video content.

This article extends social capital theory to a platform context by treating subscribers and content history as key forms of structural capital that condition subsequent engagement. Managerially, it suggests that destination marketers and creators should prioritize building subscriber bases rather than indiscriminately increasing output, and should account for the evolving engagement profile over a video's life cycle. By using broader literature, this study can serve as an anchor by providing clear definitions of engagement tiers, justifying a two-actor creator-user perspective, and demonstrating that creator-level metrics are as critical as video-level features for explaining YouTube success (Hussain et al., 2024).

Baravkar, A. et al. Sentimental Analysis of YouTube Videos

References

Baravkar, A., Jaiswal, R., Chhoriya, J., & Tekwani, B. (2021). Sentimental Analysis of YouTube Videos. *International Journal of Scientific Research in Computer Science, Engineering and Information Technology*. <https://doi.org/10.32628/cseit2172112>

Baravkar et al. (2021) propose a sentiment-enhanced ranking system for YouTube educational videos, motivated by the observation that many high-quality videos do not appear in top search positions despite strong audience appreciation (Baravkar et al., 2021). They focus on the education category, where users often watch long tutorial series and must choose carefully to avoid wasting time. The core idea is that engagement metrics alone are incomplete; rankings should also incorporate the sentiment of user comments.

The system architecture has four main modules: (1) fetching video data via the YouTube API, creating and training a machine learning sentiment model, sorting videos based on multiple parameters, and integrating the logic into a Flask-based web application. For each user query, metadata (likes, dislikes, views, comments) for the top 50 educational videos, and up to 100 comments per video are retrieved through the API, avoiding the latency of webscraping.

Sentiment classification uses a logistic regression model trained on approximately 300,000 Amazon mobile product reviews, treating ratings of 4-5 as positive and discarding neutral (rating 3) reviews to sharpen the positive class. Text is vectorized with n-grams (2-3) and minimum document frequency of 5, achieving about 90-95% accuracy on the review dataset. Each video receives a sentiment-based score derived from its top comments; this is combined with likes, comment count, views, and like-to-dislike ratio in a custom sorting function to re-rank results. A web interface displays the top nine educational videos with graphical sentiment statistics, aiming to provide students with more reliable, quality-filtered search results. Though this study gives a practical framework that utilizes the comment sentiment and engagement metrics to refine ranking of YouTube educational content, it does not directly model deeper content attributes related to virality.

Koh et al. Exploration of relation between visual attributes of thumbnails and the view-through of videos: The case of branded video content

The Koh paper sits within a growing body of work on predicting and explaining online video popularity, especially for YouTube and micro-video platforms. A core theme in this literature is that engagement is multi-dimensional: views, likes, comments, and sentiment each capture different aspects of user response, and no single factor fully explains virality.

One stream focuses on comment sentiment and basic engagement statistics. Sentiment analysis of YouTube comments shows that combining comment polarity with likes, views, and subscriptions can be used to rank educational videos so that high-quality content surfaces more effectively in search results, rather than just clickbait. Similar sentiment-based approaches on tourism videos find that subscriber counts strongly increase engagement (views, likes, comments), while very frequent uploading can actually reduce per-video engagement. Most user comments are positive, and qualitative engagement has grown over time. These studies motivate using both structural metrics (subscribers, uploads, age) and affective signals (sentiment) as predictors.

A second, technically oriented stream develops machine-learning models for micro-video popularity. Low-rank multi-view embedding methods treat popularity prediction as a regression problem, integrating heterogeneous feature sets (visual, social, temporal) into a common subspace. Transductive low-rank multi-view regression (TLRMVR) learns view-specific projections constrained to share a low-rank structure, with multi-graph regularization to improve generalization. This approach significantly outperforms strong baselines such as SVR and other multi-view methods on public micro-video datasets. Such results provide a methodological foundation for any Koh-style model that fuses multiple signals.

A third line emphasizes content and visual properties. For YouTube trending videos, combining metadata, engagement ratios, sentiment analysis (TextBlob), and topic modeling (LDA) with linear regression and XGBoost shows that both metadata and audience sentiment significantly

explain variance in view counts. Other work finds that visual memorability and atypicality can independently drive virality and commenting behavior, beyond emotional tone. Finally, educational YouTube research indicates that video length alone is not a reliable driver of engagement, highlighting the importance of content structure and context over simple duration rules.

Koh, Byungwan, and Fuquan Cui. 2022. "An Exploration of the Relation Between the Visual Attributes of Thumbnails and the View-Through of Videos: The Case of Branded Video Content." *Decision Support Systems* 160: 113820. <https://doi.org/10.1016/j.dss.2022.113820>.

Eddie's sources-

YouTube Trending Video Analysis Using Natural Language Processing and Machine Learning

Endait, Ojaswi, Mahalakshmi Iyer, Urvi Somashetti, Komal Zoting, and Dhananjay Bhagat. "YouTube Trending Video Analysis Using Natural Language Processing and Machine Learning." *2025 International Conference on Sustainability, Innovation & Technology (ICSIT)*, IEEE, 2025, 1–5. <https://doi.org/10.1109/ICSIT65336.2025.11295060>.

This study examines the factors that influence YouTube trending video popularity by combining natural language processing techniques with supervised machine learning models. Rather than treating video success as a purely algorithmic outcome, the authors focus on how textual metadata—such as video titles, descriptions, and tags—contributes to engagement metrics including views, likes, and comments.

Using a publicly available YouTube trending dataset, the authors extract linguistic features through NLP methods such as tokenization, term frequency–inverse document frequency (TF-IDF), and sentiment-related indicators. These features are then incorporated into multiple machine learning models, including linear regression and tree-based classifiers, to predict whether a video will appear in the trending category. Model performance is evaluated using standard accuracy and classification metrics, demonstrating that textual features provide meaningful predictive power when combined with basic metadata.

The findings suggest that language structure, keyword usage, and semantic patterns play a significant role in determining video visibility and audience engagement on YouTube. Videos with concise, sentiment-aware, and semantically focused titles and descriptions are more likely to attract attention and sustain interaction. This supports the broader argument that content presentation, rather than content topic alone, is a critical driver of popularity in social media platforms.

From a data analytics perspective, this paper is particularly relevant to projects focused on YouTube trend prediction and engagement modeling. The methodology aligns well with end-to-end machine learning pipelines, including feature extraction, model training, and evaluation. While the study primarily relies on textual data, its approach can be extended by integrating visual features, temporal engagement signals, and user interaction metrics. Overall, this work provides a practical and accessible example of how NLP and machine learning techniques can be applied to real-world social media datasets to understand and predict online content popularity.

Low-Rank Multi-View Embedding Learning for Micro-Video Popularity Prediction

Jing, Peiguang, Yuting Su, Liqiang Nie, Xu Bai, Jing Liu, and Meng Wang. 2018. "Low-Rank Multi-View Embedding Learning for Micro-Video Popularity Prediction." *IEEE Transactions on Knowledge and Data Engineering* 30 (8): 1519–1532.
<https://doi.org/10.1109/TKDE.2017.2785784>

This article addresses the problem of **micro-video popularity prediction** by proposing a low-rank multi-view embedding learning framework that integrates heterogeneous information sources commonly observed in social media platforms. The authors argue that video popularity is not driven by a single factor, but rather emerges from the interaction of multiple modalities, including visual features, textual metadata, and social-contextual signals. Traditional single-view or naive feature-concatenation approaches often fail to capture these interactions effectively and are susceptible to noise and redundancy.

To address these challenges, the study introduces a tensor-based low-rank multi-view representation model (TLRMVR). Each feature modality is treated as an independent view, and a shared latent representation is learned through low-rank constraints that preserve essential cross-view structures while suppressing irrelevant variation. This design enables the model to balance information sharing across views with view-specific expressiveness. The learned embedding is then used within a regression framework to predict video popularity, evaluated using normalized mean squared error (nMSE).

Empirical results on real-world micro-video datasets demonstrate that the proposed approach consistently outperforms several baseline and state-of-the-art models, including single-view regression and traditional multi-view learning methods. The performance gains highlight the importance of structured feature fusion and regularization when modeling complex social media phenomena. Additionally, the authors provide ablation studies that further validate the contribution of each modality to overall predictive performance.

From a broader data science perspective, this work is highly relevant to media analytics and machine learning–based trend prediction. Although the experiments focus on micro-video platforms, the methodological framework generalizes well to other large-scale video-sharing environments, such as YouTube. Many of the features used in this study—content descriptors, metadata, and engagement-related signals—are directly available in publicly accessible datasets. As a result, this paper offers both conceptual guidance and practical inspiration for applying advanced representation learning techniques to social media popularity forecasting tasks.

Overall, this article provides a rigorous and well-structured example of how multi-view learning and regularization can be applied to noisy, high-dimensional social media data. Its emphasis on feature interactions, dimensionality reduction, and predictive evaluation aligns closely with modern machine learning workflows and makes it a strong foundational reference for projects involving video analytics and popularity prediction.

Machine Learning Pipelines for Deepfake Text Detection on Social Media

Gaur, Adamya, Sanjay Kumar Singh, and Pranshu Saxena, (2024). "Performance Analysis of Deepfake Text Detection Techniques on Social-Media." *International Conference on Distributed Computing and Optimization Techniques (ICDCOT)*, IEEE, 2024, 1–6.
<https://doi.org/10.1109/ICDCOT61034.2024.10515626>

The use of natural language processing has created new issues for social media users; specifically the creation of "deepfake" and "spurious" text is spreading rapidly. A recent study by Gaur et al. (2024) explores the issue of detecting deepfake texts through the lens of machine-generated content produced by social bots on X (formerly Twitter). The researchers illustrate the growth in complexity of automated text generation systems like RNNs and GPT-2. GPT-2 produces text that is increasingly indistinguishable from human-created text.

In terms of evaluating their models' ability to detect deepfake texts, Gaur et al. (2024) used the publicly available TweepFake dataset, which contains 25,572 labeled tweets. The data was cleaned using typical preprocessing steps for text cleaning, such as removing missing values, normalizing the case of the data, converting all label information into numerical format, and removing stopwords, punctuation, and URLs. Gaur et al. (2024) tested a number of different models, ranging from traditional machine learning models (such as Logistic Regression, XGBoost, and AdaBoost) that were trained on both Term Frequency (TF) and Term Frequency-Inverse Document Frequency (TF-IDF) features to more advanced deep learning models (such as DNN, CNN, and Bi-LSTM), and finally to more advanced transformer-based models (including BERT, DistilBERT, and RoBERTA).

Results of the experiments indicate that the most effective approach for identifying deepfake texts was the use of transformer-based models. Specifically, BERT achieved an accuracy rate of 89.17%, the highest precision, and F1 score. While DistilBERT did not achieve the same level of performance as BERT, it achieved comparable levels of performance while requiring less computational resources than BERT. Among the evaluated deep learning models, Bi-LSTM performed the best. The results of Gaur et al.'s study suggest that the use of transformer-based models will likely prove to be the most effective models for large-scale text classification tasks.

While the specific focus of Gaur et al.'s study was not similar to that of previous studies focused on predicting user engagement or popularity, the methodological approach they used can be seen as being applicable to other prediction tasks of social media research. For example, Gaur et al.'s use of supervised machine learning methods to process large social media datasets, including data preprocessing, feature engineering, comparing multiple models, and evaluating model performance, demonstrates a systematic methodology that can be applied to other large social media datasets when adapting feature sets and prediction targets regardless of whether the prediction target is engagement, popularity, or some other measure of user behavior. Therefore, the study demonstrates how a generalizable methodology can provide contextually-related support for the application of machine learning across various types of social media analysis.